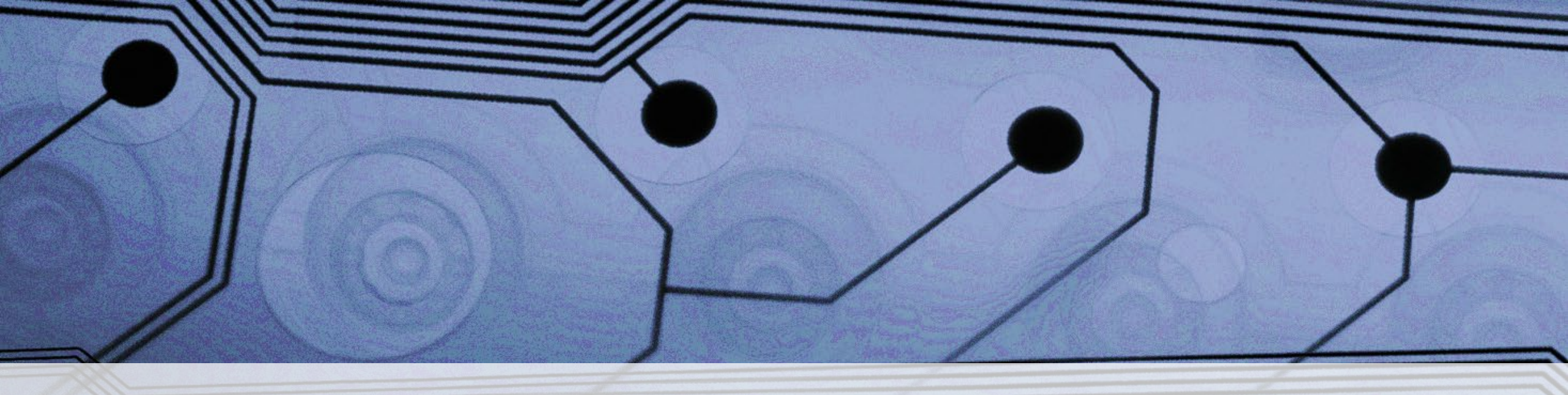
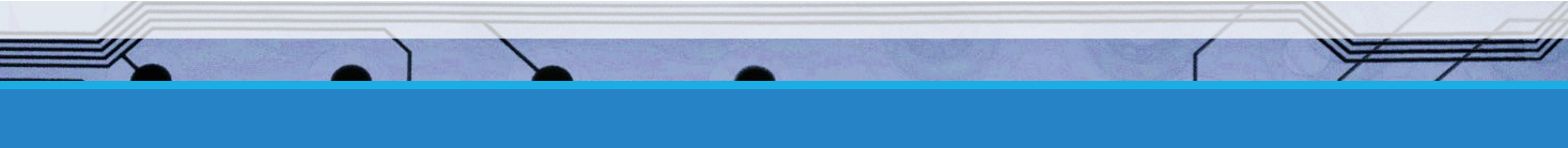




Hands-on PCB Design and Electronic Prototyping with Autodesk Fusion

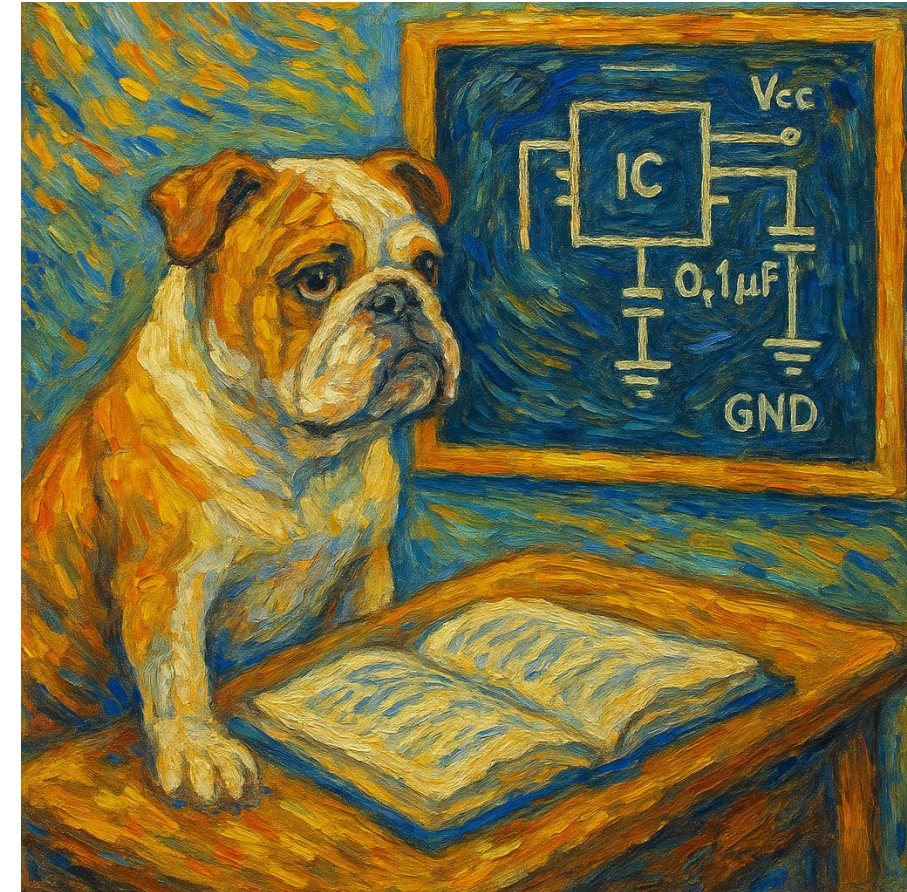
SUMMER ENRICHMENT- LYLES COLLEGE OF ENGINEERING

REVISION 1.0



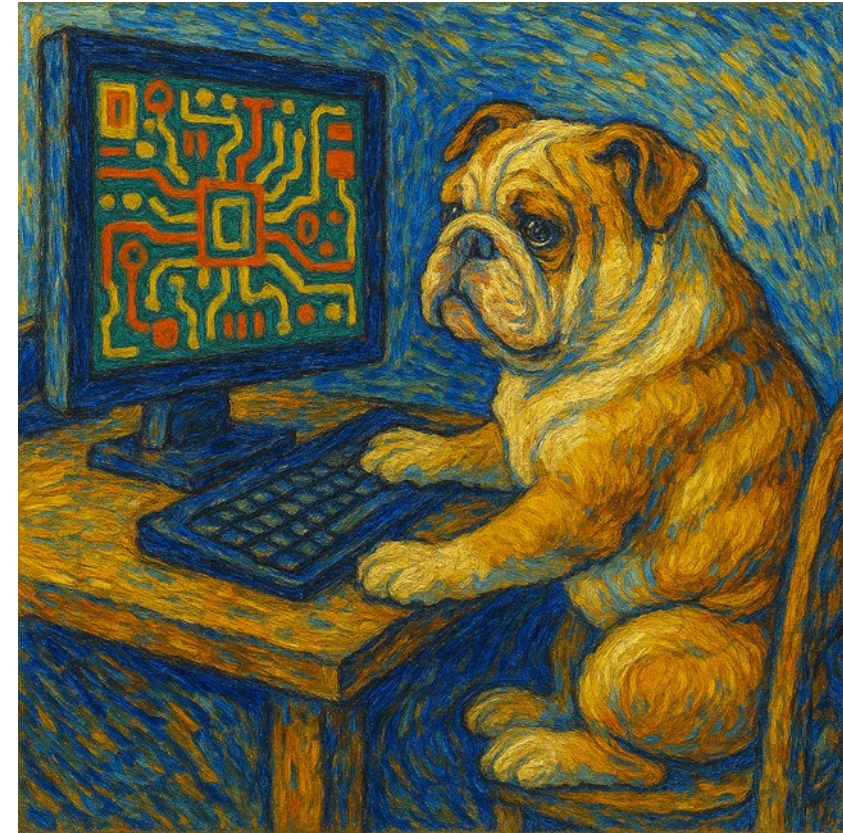
Day 4: Review

- PCB Layout Essentials (Part 1)
 - Transitioning from Schematic to PCB
 - Generating PCB from schematic
 - Understanding PCB workspace and tools
 - Component Placement
 - Placement strategies (signal integrity, EMI/EMC, thermal considerations)
 - Floor-planning
 - Hands-on Activity
 - Component placement for a simple project PCB



Day 5: PCB Layout Essentials (Part 2)

- Routing Basics
 - Trace routing (width, spacing, clearance rules)
 - Via usage and types
- Design Considerations
 - Ground and power planes, copper pours, and polygon regions
- Hands-on Activity
 - SMT Voltage Regulator Schematic/PCB



Surface Mount vs. Through Hole

- **SMT (Surface-Mount Technology)**

- Components mounted directly on PCB surface
- Smaller, lighter, more compact designs
- Ideal for high-density, high-speed circuits
- Automated placement = faster & cheaper for mass production
- Lower mechanical strength



Surface Mount vs. Through Hole

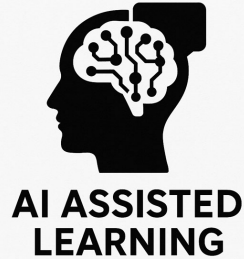
- **Through-Hole Technology**

- Component leads inserted through PCB holes and soldered
- Strong mechanical bonds – better for high-stress applications
- Easier for prototyping and manual soldering
- Bulky – limits component density
- Slower, costlier manufacturing





AI Minute



-
- Please type the following prompts into the AI Tool of your choice, Chat GPT, Gemini, Grok, Copilot, etc...
 - **From a design engineer's perspective, contrast and compare SMT components use in a PCB Assembly vs through hole components use in a PCB assembly.**

Polygons & Copper Pours: Purpose and Use in PCB Design

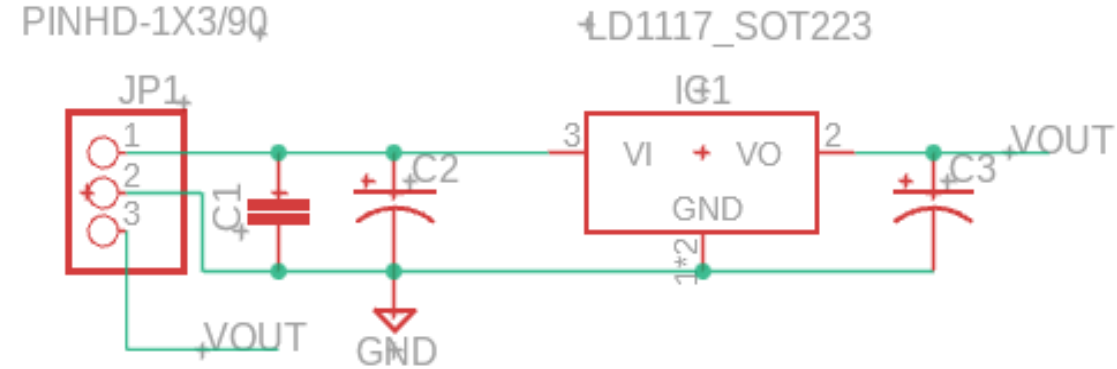
- What are Polygons and Copper Pours?
 - Polygons: Closed shapes defined in PCB layout to represent large copper areas.
 - Copper Pour: A fill of copper within a polygon—used for grounding, power planes, or thermal management.
- Why Use Them?
 - Improved Grounding: – Reduces ground impedance and improves signal integrity.
 - Thermal Management: – Spreads heat away from high-power components.
 - Current Handling: – Provides low-resistance paths for high-current traces.
 - EMI Reduction: – Acts as a shield to minimize electromagnetic interference.
 - Efficient Material Use: – Minimizes copper waste and helps with PCB etching balance.

Modify Amplifier Schematic- Hands On

- Open your Amplifier Schematic
 - Replace the decoupling capacitor with
 - CAP-THRU-2.54MM- Tinkercad- Capacitor Library
 - Save schematic
 - Review PCB
 - Update if necessary

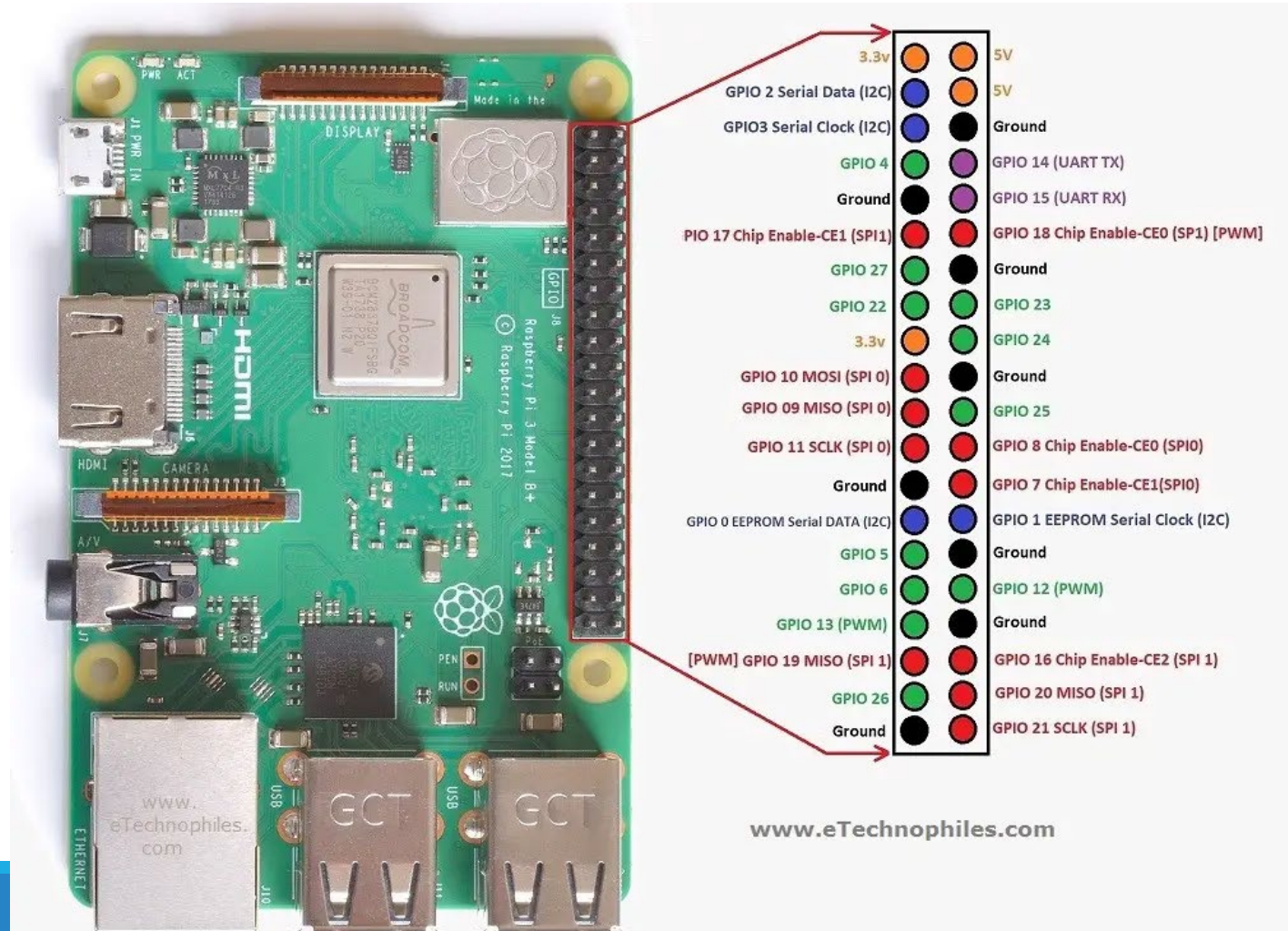
Hands-On SMT Project

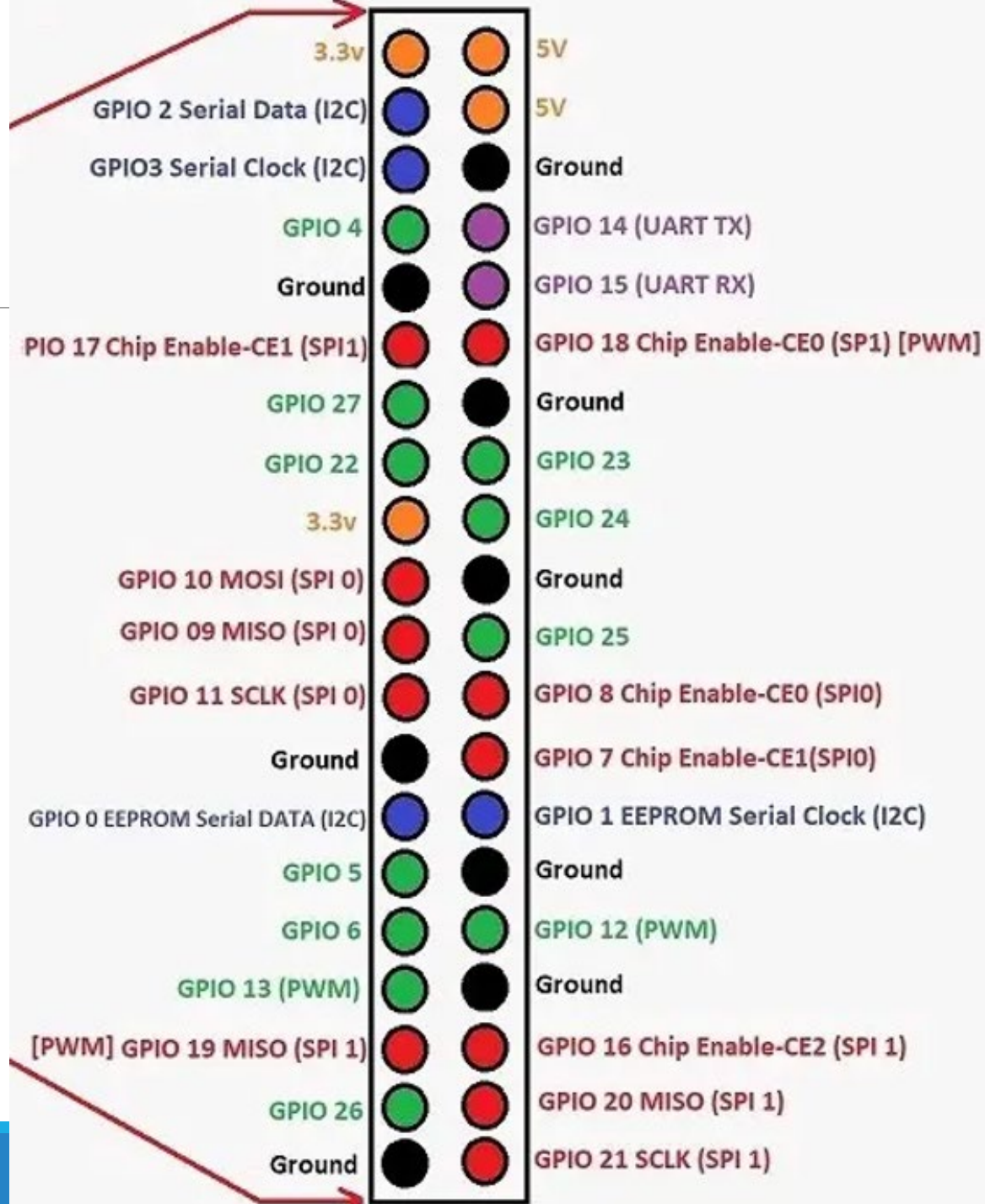
- SMT LDO Project
 - Create a new electrical project in Fusion
 - Create the schematic
 - Components:
 - IC1: LD1117, SOT23 package- AutoDesk IC-Power Management Library
 - JP1: PINHD-1x3/90- AutoDesk- Connector Library
 - C1: CAP-THRU-2.54MM- Tinkercad- Capacitor Library
 - C2, C3: ECAP-8.5MM- AutoDesk- C-POL-US Library
 - Save schematic
 - Create PCB
 - Be sure to use ground planes and copper pours where practical



Raspberry Pi Pinout- Create Library Component- PCB Footprint

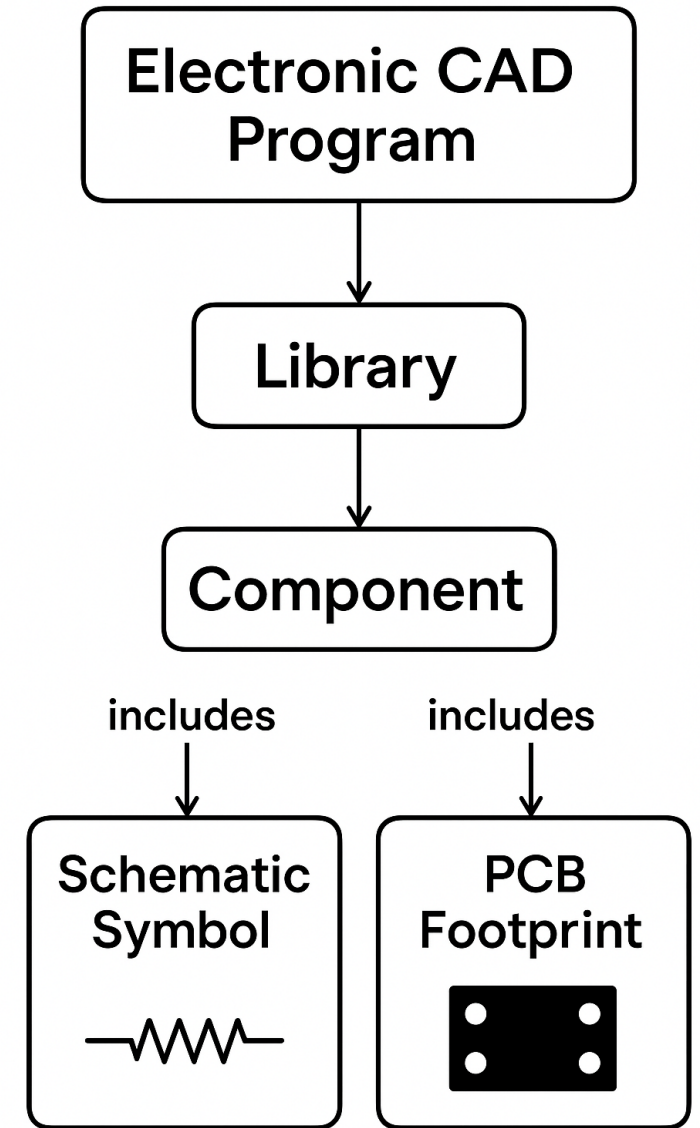
- Component Creation
 - Create Schematic Symbol
 - Create PCB Footprint
 - Combine Footprint and Symbol into a Component
- Create a new project
- Place the newly created Raspberry Pi I/O connector component on the schematic and run ERC
- Switch to PCB and place the component on the PCB
- Adjust the PCB outline to be a more appropriate size for the connector





CAD Library Elements

- Libraries
 - Libraries contain components
- Components
 - Components include:
 - A Schematic Symbol
 - A PCB Footprint
 - A 3D Model



Questions?

